Joshua Tlatelpa-Agustin
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EDUCATION

PhD Computer Science **Sep 2026 -**
The University of British Columbia *Vancouver, BC*

MS Computer Science **GPA: 4.0/4.0** **2025 - May 2026**
The University of Utah *Salt Lake City, Utah*

- *Thesis: Towards Hash Tables Operating at the Speed of DRAM*
- *Committee: Anton Burtsev (chair), John Regehr, Vijay Nagarajan*

BS Computer Science **2022 - 2024**
The University of Utah *Salt Lake City, Utah*

Guest Student **2021 - 2023**
Salt Lake Community College *Salt Lake City, Utah*

Graduate Coursework: Adv. Compilers, Adv. Operating Systems, Interactive Computer Graphics, Adv. Operating Systems II, Computer Architecture, Software Verification, HC Data Management, Adv. Computer Architecture

Undergrad Coursework: Computer Architecture, Algorithms & Data Structures, Discrete Structures, Object Oriented Programming, Models of Computation, Probability & Statistics, Software Practice 1 & 2, Computer Systems, Compilers, OS, Database Systems, Computer Graphics, Visualization for Data Science, Algorithms

SKILLS

- *Programming Languages:* C, C++, Rust, Python, Javascript, ASM (x86), Java, C#, R
- *Additional Skills:* Performance profiling, MySQL, Spanish (fluent)

PUBLICATIONS

- Zhaofeng Li, Jerry Zhang, Joshua Tlatelpa-Agustin, Xiangdong Chen, and Anton Burtsev. Understanding the Security Impact of CHERI on the Operating System Kernel. *In Proceedings of the Annual Computer Security Applications Conference (ACSAC)*, December 2025.
- Jerry Zhang, Joshua Tlatelpa-Agustin, and Anton Burtsev. DRAMHiTv2: Towards the fastest hash table operating at the speed of DRAM. *Submitted to the European Conference on Computer Systems (EuroSys)*, April 2026. (Title modified for anonymity.)

RESEARCH

CHERI *Accepted*

- Study of CHERI's (Capability Hardware Enhanced RISC Instructions) impact on OS kernels. Analysis of 440 Linux and FreeBSD kernel CVEs revealed that CHERI could prevent 60% of vulnerabilities, including the majority of critical privilege escalations.

DRAMHiTv2 *Submitted*

- A next-generation in-memory hash table that reaches hardware bandwidth limits and maximizes operational throughput through a multi-level prefetching scheme, a compute–memory–aware table layout, and a conflict-resolution strategy optimized for memory bandwidth utilization.

PROJECTS

JPL Compiler (class solo project)

- Developed a compiler for the JPL programming language. The project included a lexer, parser, type checker, simple optimizations, and assembly generation. Spec: https://github.com/utah-cs4470-sp23/class/blob/2023/spec.md
- *Technologies:* C++ (~12,000 lines), x86-64 asm

BF Compiler/Interpreter/Profiler (class solo projects *[graduate level]*)

- Developed an interpreter and compiler for BF, a minimalist, Turing-complete esoteric language. The compiler includes optimization passes for loop elimination and utilizes vector instructions in assembly generation to accelerate memory seeking. Spec: https://www2.gvsu.edu/miljours/bf.html
- *Technologies:* C++, x86-64 asm, SIMD, LLVM

Learning Management System (class project, group of two)

- Multi-phase project to develop a learning management system (LMS) resembling Canvas. Involved designing a database, creating SQL tables, building a web server, and online deployment.
- *Technologies:* C#, MySQL

OpenGL Project(s) (class solo projects *[graduate level]*)

- Wrote various real-time interactive graphics applications each utilizing one of the following techniques: transformations, shading, textures, render buffers, environment mapping, shadow mapping, tessellation, and instancing.
- *Technologies:* C++, OpenGL, GLFW, GLEW

HONORS AND AWARDS

UBC BPOC Graduate Excellence Award (2026)

The award is intended to recognize excellence in research and scholarly performance by BPOC (Black and People of Colour) graduate students and to address systemic barriers they experience in their training.

- $1,500 CAD

UBC Computer Science Merit Scholar (CS Merit) (2026)

The UBC Department of Computer Science is offering CS Merit Awards to outstanding incoming international and domestic students. Students awarded a CS Merit Award are guaranteed award-level funding in their first year at UBC.

- $8,000 CAD

Jerry Taylor Scholarship (2023)

This scholarship is designed to support second year students who have demonstrated course & character in response to a major challenge in life, or in pursuit of their education.

- $2,500 USD